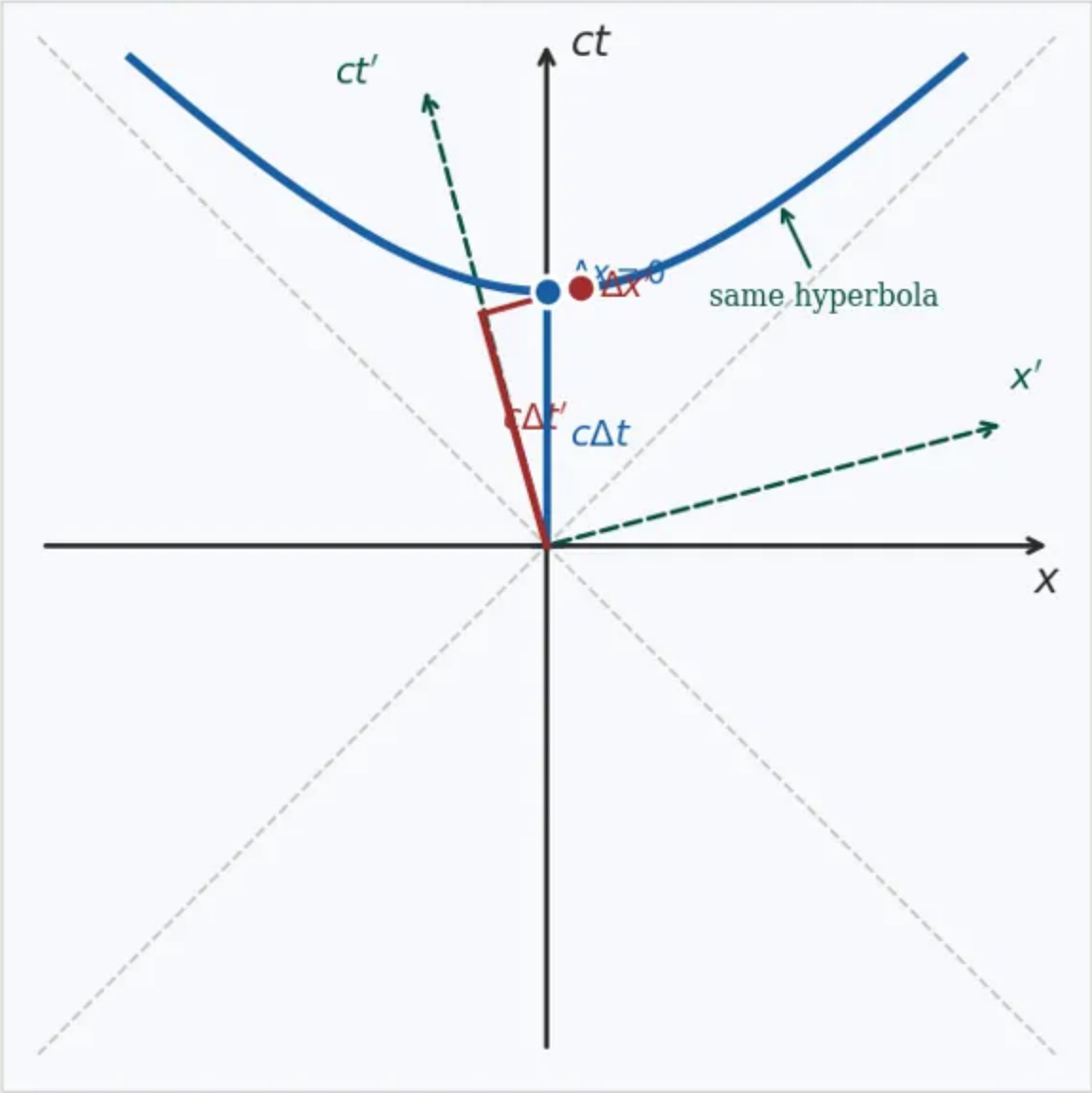
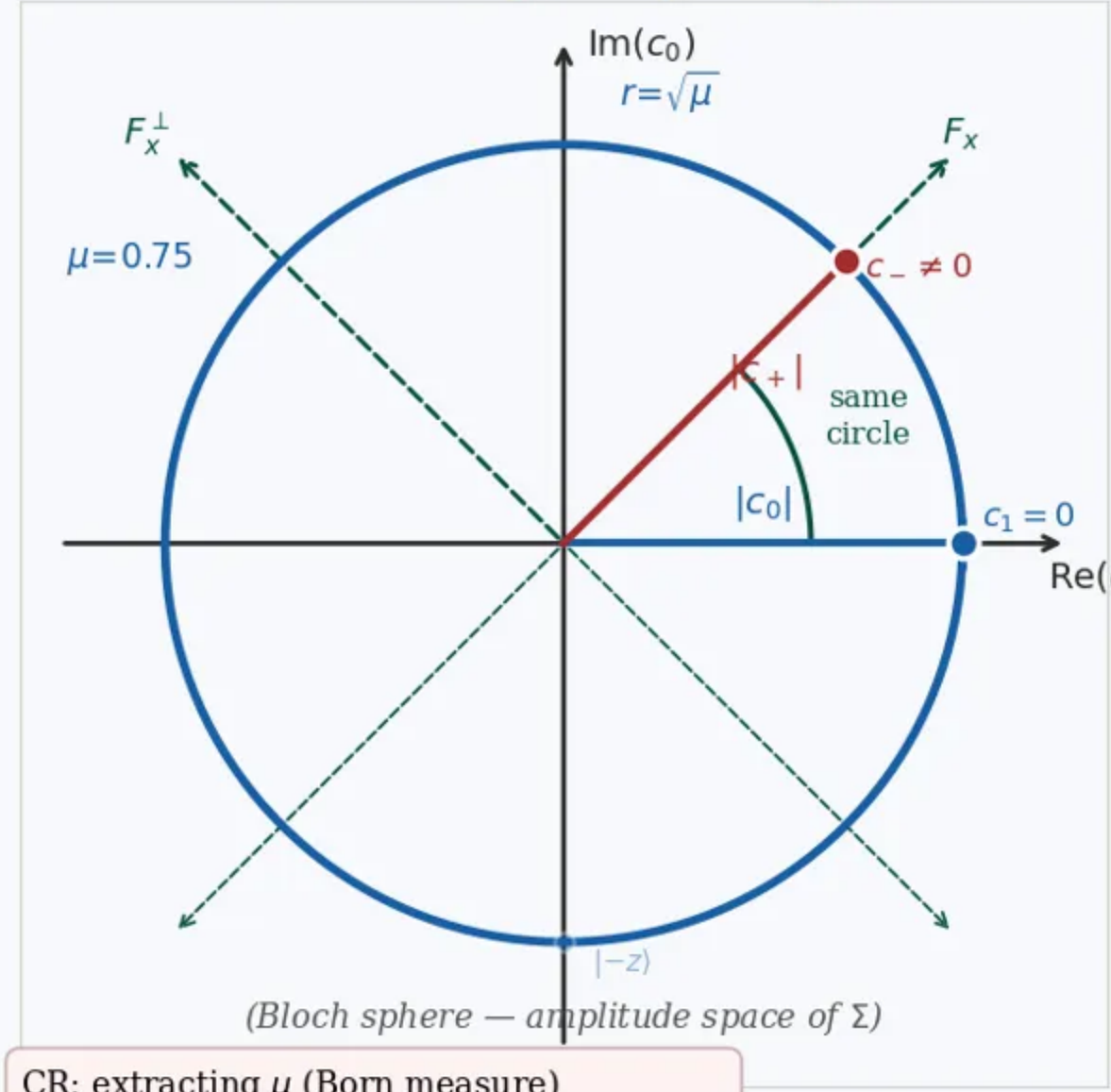
The Born rule as frame invariant: ds^2 in SR $\longleftrightarrow$ $\mu = |c_i|^2$ in CR

Special Relativity — Minkowski diagram



Coherence Relativity — amplitude plane of Σ



SR: extracting ds^2

Frame S: $(c\Delta t,\ \Delta x) = (53,\ 0)$
$ds^2 = -53^2 + 0^2 = \mathbf{-2809}$

Frame S': $(c\Delta t',\ \Delta x') = (47,\ 24)$
$ds^2 = -47^2 + 24^2 = \mathbf{-2809}$

$\Rightarrow$ ds^2 identical in both frames

CR: extracting μ (Born measure)

Frame F_z: $(\mathrm{Re}\, c_0,\ \mathrm{Im}\, c_0) = (0.866,\ 0)$
$\mu = |c_0|^2 = 0.866^2 = \mathbf{0.75}$

Frame F_x: $(c_+,\ c_-) = (0.612,\ 0.612)$
$|c_+|^2 = 0.375 \neq 0.75$ (frame-dependent)

But $r = \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2} = 0.866$
$\Rightarrow$ $\mu = r^2 = \mathbf{0.75}$ — invariant on Born circle